

## Problem Solving

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1. It is proposed to fit the Poisson distribution to the following data:

| x         | 0  | 1  | 2  | 3  | $3 < x$ |
|-----------|----|----|----|----|---------|
| Frequency | 20 | 40 | 16 | 18 | 6       |

- (a) Compute the corresponding chi-square goodness-of-fit statistic.
- (b) How many degrees of freedom are associated with this chi-square?
- (c) Do these data result in the rejection of the Poisson model at the  $\alpha = 0.05$  significance level?



2. Determine a method to generate random observations for the extreme-valued pdf that is given by

$$f(x) = \exp\{x - e^x\}, \quad -\infty < x < \infty.$$

3. Consider the pdf

$$f(x) = \begin{cases} \beta x^{\beta-1} & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

for  $\beta > 1$ .

- (a) Use the CDF technique to generate an observation from this pdf.
- (b) Use the accept-reject algorithm to generate an observation from this pdf.

